

1. Title Slide

Explainability for Crypto Trading

Interpretable AI for Financial Decision-Making

2. TL;DR

Crypto markets are volatile and opaque.

Explainability improves trust in algorithmic trading.

Results: Insights into which indicators drive Buy/Sell decisions.

3. Motivation

Black-box crypto models lack transparency.

Retail investors need clearer reasoning behind decisions.

Explainable AI supports safe deployment.

4. Literature Review

SHAP, LIME, Integrated Gradients, Attention methods.

Explainable Reinforcement Learning frameworks.

Key challenge: Explanation evaluation.

5. Problem Definition

Goal: Understand WHY the model buys/sells.

Challenge: Non-stationary market behavior.

Need interpretable, stable explanations.

6. Approach Overview

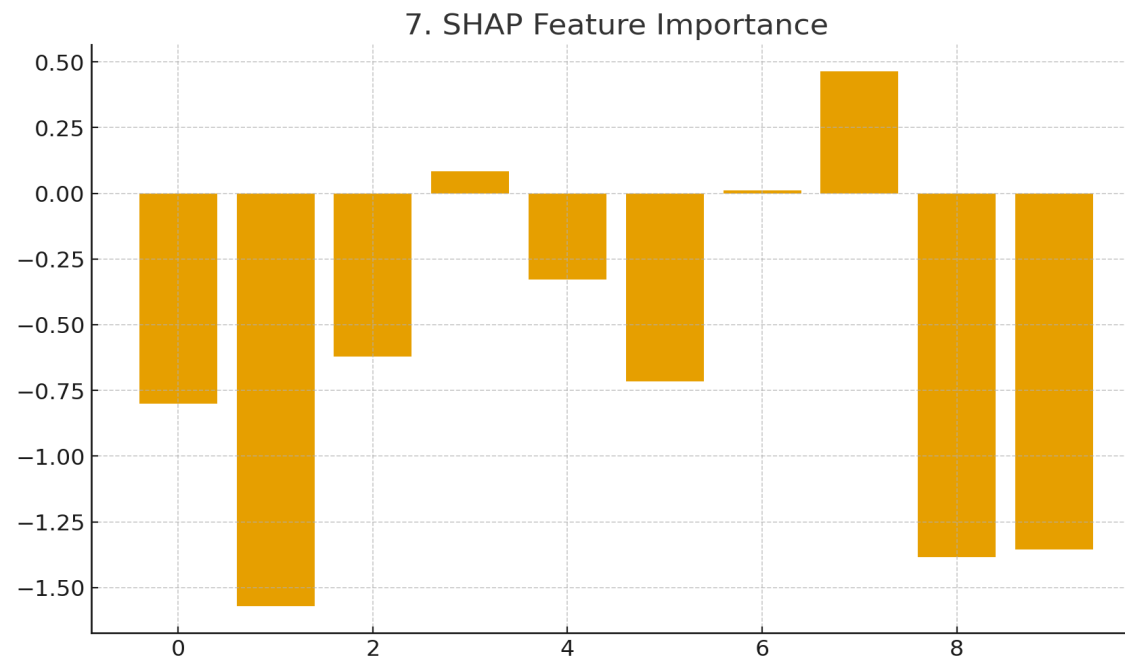
Model: LSTM / Transformer / RL Agent.

Data: BTC OHLCV + indicators.

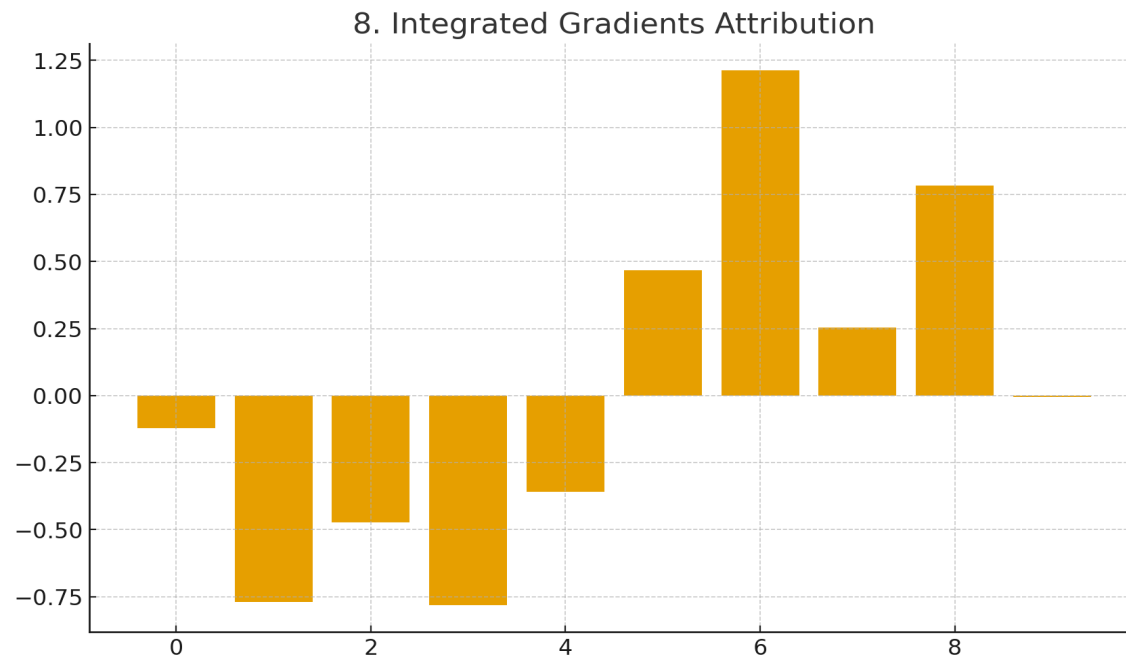
XAI Tools: SHAP, IG, Attention.

Evaluation: Stability, consistency.

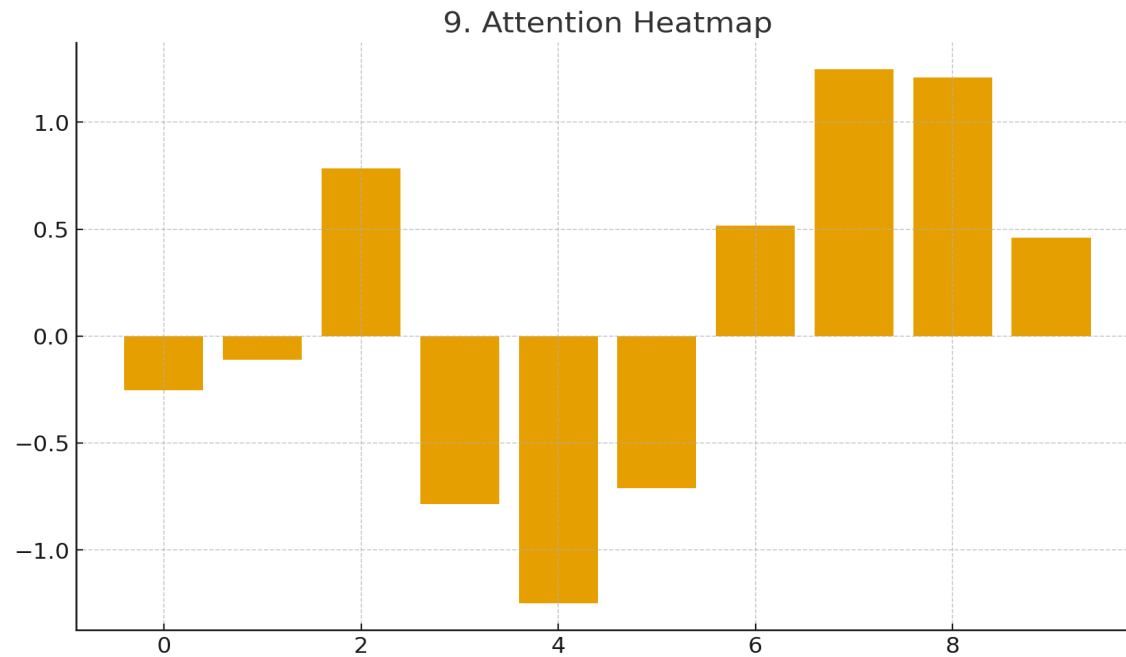
7. SHAP Feature Importance



8. Integrated Gradients Attribution



9. Attention Heatmap



10. Results Summary

Buy signals: RSI, MACD, EMA momentum patterns.

Sell signals: volatility spikes & reversals.

Explanations match financial intuition.

11. Ethical Considerations

Explanations may mislead if misinterpreted.

High volatility reduces explanation reliability.

Avoid overtrust in automated systems.

12. Conclusion & Future Work

Explainability increases transparency & trust.

Future: sentiment signals, counterfactuals, user studies.

Better evaluation metrics needed.